Web Analytics Analysis Project

Overview

This project analyses website activity data from **datadb.com** to understand how users interact with the site and which variables are associated with key engagement outcomes. The dataset includes measures such as **Bounces, Continent, Source group, Time on page, Unique pageviews, Visits, and Exits**.

Objective

The main goal of the analysis was to:

- summarize the dataset for a basic understanding
- test whether Unique pageviews depend on Visits
- identify factors affecting Exits
- identify factors affecting Time on Page
- identify factors affecting Bounces

Tools Used

- **R**
- readxl
- ggplot2
- car

Workflow

1. Data Import and Preparation

The dataset was imported from an Excel file into R and column names were cleaned using make.names() to ensure they could be used safely in models and plots.

2. Data Summarization

The data was explored using:

- str()
- summary()
- custom numeric summaries

This helped provide an overview of the dataset and the type of values stored in each variable.

3. Unique Pageviews vs Visits

A Pearson correlation test and a simple linear regression were used to determine whether Unique Pageviews depends on Visits. The results showed a strong positive relationship, meaning that higher visits were associated with higher unique pageviews.

4. Factors Affecting Exits

A multiple linear regression model was used to identify predictors of **Exits**.
The analysis showed that:

- **Bounces** had a strong positive effect,
- **Unique Pageviews** also had a strong positive effect,
- **Visits** had a smaller but significant positive effect,
- several **Source group** categories were significant,
- most **Continent** variables were not significant.

5. Factors Affecting Time on Page

A multiple linear regression model was built to examine variables influencing **Time on Page**.
The significant predictors were:

- Bounces: strong negative effect
- Unique Pageviews: positive effect
- Visits: strongest positive effect
- Exits: positive effect
- South America (SA): significant continent category.

The model explained a smaller portion of variation in Time on Page, so its overall fit was relatively weak, although some predictors were still statistically significant.

6. Factors Affecting Bounces

A multiple linear regression model was used to identify variables affecting Bounces.
The results showed that:

- Time on Page had a strong negative effect
- Unique Pageviews had a negative effect
- Visits had a strong positive effect
- Exits had a strong positive effect

- several Source group categories were significant
- most Continent variables were not significant

Key Findings

- Visits and Unique Pageviews are strongly related.
- Bounces is one of the most important predictors across the models.
- Time on Page decreases when bounce behaviour increases.
- Visits and Exits are strong positive predictors in the Bounces model.
- Source group is an important factor in understanding user behaviour.
- Continent is generally less important, with only a few categories showing significance.

Conclusion

The project shows that user engagement on the website is influenced by a combination of behavioural and traffic-source variables. The strongest relationships were found in the models for Exits and Bounces, while Time on Page was more difficult to explain using the available variables. Overall, the analysis provides useful insight into how users interact with the website and which factors are most associated with exits, bounce behaviour, and page engagement.